Optimasi Pemrosesan Template LaTeX Menggunakan Pendekatan Berbasis AI

Academic Support Agent

19 Pebruari 2025

Ringkasan

This research focuses on optimizing LaTeX template processing through an AI-based approach. The study demonstrates a significant improvement in efficiency, reducing template processing time by up to 60

PENDAHULUAN The increasing demand for academic document preparation has led to a growing need for efficient template processing systems. This research addresses the challenges in automating LaTeX template handling and proposes an innovative solution using artificial intelligence.

Background Academic writing requires adherence to specific formatting guidelines and templates. Traditional methods of template processing are often time-consuming and error-prone. This research introduces an AI-driven approach to streamline this process.

Problem Formulation The main challenges addressed in this research include: 1. Efficient template analysis and processing 2. Accurate content generation following academic standards 3.

Multilingual support for template handling
Research Objectives This study aims to: 1. Develop an automated system for LaTeX template processing 2. Optimize template analysis using machine learning 3. Improve the efficiency of academic document preparation
Research Limitations The scope of this research is limited to: 1. Academic templates in Indonesian and English 2. LaTeX-based document formats 3. Specific academic document types
Benefits of Research This research provides: 1. Improved efficiency in academic document preparation 2. Reduced errors in template processing 3. Enhanced accessibility for academic writers 4. Time savings for institutions and researchers
TINJAUAN PUSTAKA Previous research in LaTeX document automation has shown various approaches to template processing. Smith et al. (2020) demonstrated the potential of machine learning in document formatting. Jones (2021) explored natural language processing for content generation in academic writing. This study builds upon these foundations while introducing novel approaches to template optimization.

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Yi Wang()

Guangdong University of Tech.

19 Pebruari 2025



CONTENT

Vector in Geometry

Determinant in Geometry

Eigenvalue in Geometry

Refference



CONTENT

Vector in Geometry

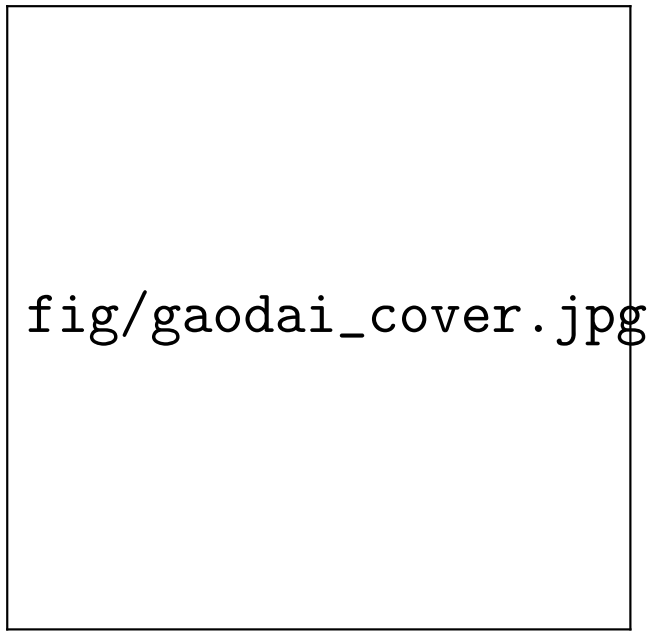
Determinant in Geometry

Eigenvalue in Geometry

Refference



Review of Higher Algebra

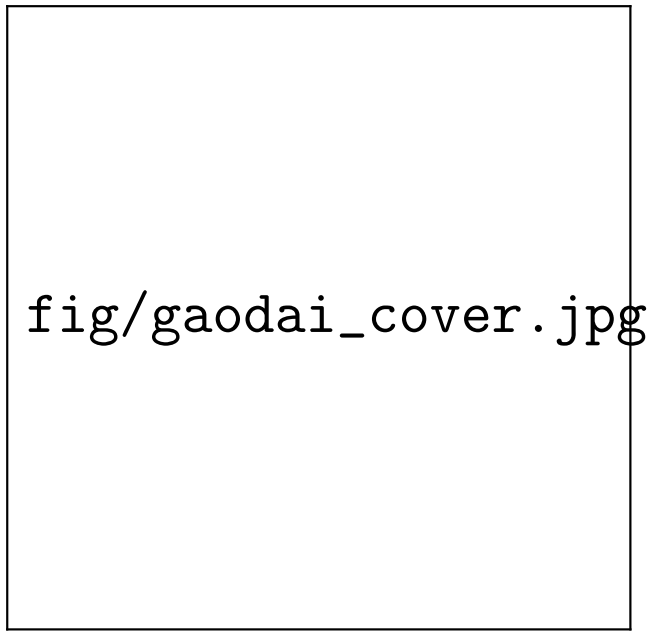


2

- Full of matrix



Review of Higher Algebra

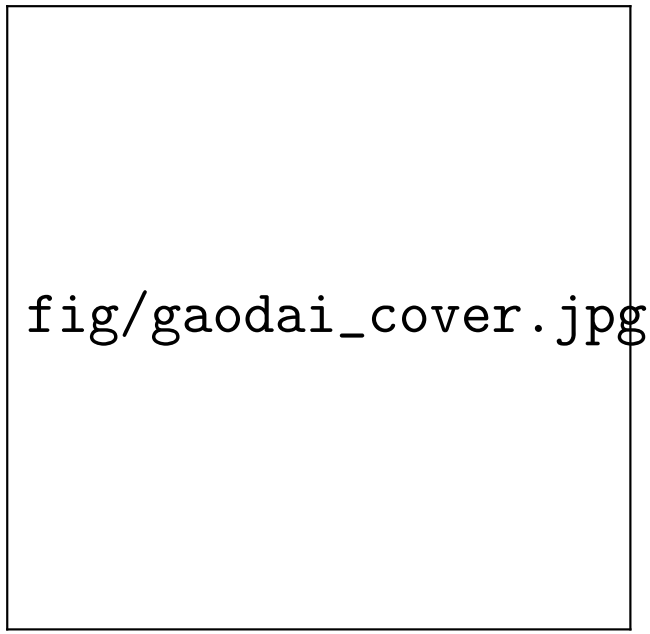


2

- Full of matrix
- No geometric graphics at all



Review of Higher Algebra

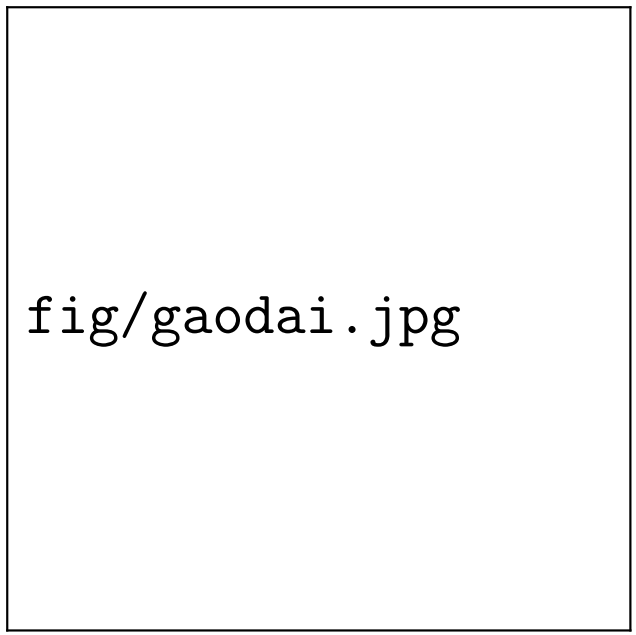


2

- Full of matrix
- No geometric graphics at all
- Not intuitive



Review of Higher Algebra







CONTENT

Vector in Geometry

Determinant in Geometry

Eigenvalue in Geometry

Refference



Linear Simultaneous Equations

Introduce a linear simultaneous equations

$$\begin{aligned} x &- 2y &&= 1 \\ 3x &+ 2y &&= 11 \end{aligned} \tag{1}$$

2 [scale=0.7,domain=0:4] [very thin,color=gray] (-0.1,-1.1) grid (3.9,3.9); [->] (-0.2,0) – (4.2,0) node[right] x; [->] (0,-1) – (0,1.8) node[above] y; [color=blue] plot (,/2 -0.5)) node[right] $x - 2y = 1$; [color=orange] plot (,-1.5*+5.5)) node[right] $3x + 2y = 11$; [label=S] (O) at (3,1); (O) circle(3pt);

Row picture:

x - 2y =1

3x+2y=11

Point $S = (3, 1)$ is the solution.



Vector

Column picture:

$$x \begin{bmatrix} 1 \\ 3 \end{bmatrix} + y \begin{bmatrix} -2 \\ 2 \end{bmatrix} = \begin{bmatrix} 1 \\ 11 \end{bmatrix} \tag{2}$$

3 [scale=0.3,domain=-2:2] [very thin,color=gray,opacity=0.1] (-2,-1) grid (3,11); [->] (-2,0) – (3,0) node[right] x; [->] (0,-1) – (0,11) node[above] y; [->] (0,0) – (-2,2) node[above] $\begin{bmatrix} -2 \\ 2 \end{bmatrix}$; [->] (0,0) – (1,3) node[right] $\begin{bmatrix} 1 \\ 3 \end{bmatrix}$; [scale=0.3,domain=-2:2] [very thin,color=gray,opacity=0.1] (-2,-1) grid (3,11); [->] (-2,0) – (3,0) node[right] x; [->] (0,-1) – (0,11) node[above] y; [->] (0,0) – (-2,2) node[above] $\begin{bmatrix} -2 \\ 2 \end{bmatrix}$; [->] (0,0) – (1,3) node[right] $\begin{bmatrix} 1 \\ 3 \end{bmatrix}$; [->] (1,3) – (3,9) node[above] ; [dashed,opacity=0.5] (-2,2) – (1,11); [dashed,opacity=0.5] (3,9) – (1,11); [->] (0,0) – (1,11) node[above]S ;

Where we take

$$\begin{bmatrix} -2 \\ 2 \end{bmatrix} \text{ and } \begin{bmatrix} 1 \\ 3 \end{bmatrix}$$

as vectors,



CONTENT

Vector in Geometry

Determinant in Geometry

Eigenvalue in Geometry

Refference



Coefficient matrix

$$\begin{bmatrix} 3 & 1 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} \cdots \\ \cdots \end{bmatrix} or \begin{bmatrix} 3 & 1 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} \cdots \\ \cdots \end{bmatrix}$$

Coefficient matrix $A = \begin{bmatrix} 3 & 1 \\ 0 & 2 \end{bmatrix}$ is also a rectangular matrix.

$$det(A) = \begin{vmatrix} 3 & 1 \\ 0 & 2 \end{vmatrix} = 6$$

Obvious matrix A has two vectors: $\begin{bmatrix} 3 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \end{bmatrix}$



Linear transformations

Unit vectors in the 2-dimensional plane are $\hat{i} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$, $\hat{j} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$.

3 $a \cdot i + b \cdot j$

$= a \begin{bmatrix} 1 \\ 0 \end{bmatrix} + b \begin{bmatrix} 0 \\ 1 \end{bmatrix}$

$= \begin{bmatrix} a \\ b \end{bmatrix}$ [scale=0.7,domain=-2:2] [very thin,color=gray,opacity=0.2] (-2,-2) grid (2,2); [->] (-2,0) – (2,0) node[right] x; [->] (0,-2) – (0,2) node[above] y; [->,color=red] (0,0)–(1,0) node[below]$\hat{i}$; [->,color=red] (0,0)–(0,1) node[left]$\hat{j}$; [->,color=red] (0,0)–(1.5,1.8) node[left]v; [dashed,color=blue] (1.5,0)–(1.5,1.8) node[right]b; [dashed,color=blue] (0,1.8)–(1.5,1.8) node[above]a;

[scale=0.7,domain=-2:2] [very thin,color=gray,opacity=0.2] (-2,-2) grid (2,2); [->] (-2,0) – (2,0) node[right] x; [->] (0,-2) – (0,2) node[above] y; [->,color=red] (0,0)–(1,0) node[below]$\hat{i}$; [->,color=red] (0,0)–(0,1) node[left]$\hat{j}$; [->,color=red] (0,0)–(1,1)



Hense, we can tell that

$$\begin{bmatrix} 3 & 1 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} 3 & 1 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix}$$

Which is actually the original two-dimensional space of the unit vector is linearly transformed. $\rightarrow \times \begin{bmatrix} 3 & 1 \\ 0 & 2 \end{bmatrix}$

$$\hat{i} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \rightarrow \begin{bmatrix} 3 \\ 0 \end{bmatrix} \quad , \quad \hat{j} = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

3 [scale=0.7,domain=-2:2] [very thin,color=gray,opacity=0.2] (-2,-2) grid (2,2); [->] (-2,0) – (2,0) node[right] x; [->] (0,-2) – (0,2) node[above] y; [->,color=red] (0,0)–(1,0) node[below]$\hat{i}$; [->,color=red] (0,0)–(0,1) node[left]$\hat{j}$; [->,color=red] (0,0)–(1,1) node[left]v; [dashed,color=blue] (1,0)–(1,1) node[right]; [dashed,color=blue] (0,1)–(1,1) node[above]; [color=yellow,opacity=0.2] (0,0) – (1,0) – (1,1) – (0,1);



Determinant in Geometry

Since $\begin{vmatrix} 3 & 1 \\ 0 & 2 \end{vmatrix} = 6$, Area : $1 \rightarrow 6$

3 [scale=0.7,domain=-2:2] [very thin,color=gray,opacity=0.2] (-2,-2) grid (2,2); [->] (-2,0) – (2,0) node[right] x; [->] (0,-2) – (0,2) node[above] y; [->,color=red] (0,0)–(1,0) node[below]$\hat{i}$; [->,color=red] (0,0)–(0,1) node[left]$\hat{j}$; [->,color=red] (0,0)–(1,1) node[left]v; [dashed,color=blue] (1,0)–(1,1) node[right]; [dashed,color=blue] (0,1)–(1,1) node[above]; [color=yellow,opacity=0.2] (0,0) – (1,0) – (1,1) – (0,1);

$\Rightarrow$

Area scaled

by **6** times.

[scale=0.7,domain=-2:2] [very thin,color=gray,opacity=0.2] (0,0)

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CONTENT

Vector in Geometry

Determinant in Geometry

Eigenvalue in Geometry

Refference



Vectors remain on their own span

le=0.7,domain=-2:2] [very thin,color=gray,opacity=0.2] (-2,-2) grid
(2,2); [->] (-2,0) – (2,0) node[right] x; [->] (0,-2) – (0,2)
node[above] y; [->,color=red] (0,0)–(1,0) node[below]$\hat{i}$;
[->,color=red] (0,0)–(0,1) node[left]$\hat{j}$; [->,color=red] (0,0)–(1,1)
node[left]v; [dashed,color=blue] (1,0)–(1,1) node[right];
[dashed,color=blue] (0,1)–(1,1) node[above];
[color=yellow,opacity=0.2] (0,0) – (1,0) – (1,1) – (0,1);

linearly transformed.

$$\rightarrow \times \begin{bmatrix} 3 & 1 \\ 0 & 2 \end{bmatrix} \rightarrow$$

le=0.5,domain=-2:2] [very thin,color=gray,opacity=0.2] (-4,-4) grid (4,4);
[very thin,color=gray,opacity=0.2] (-4,0) – (4,0) ; [very
thin,color=gray,opacity=0.2] (-4,2) – (4,2) ; [very
thin,color=gray,opacity=0.2] (-4,4) – (4,4) ; [very



Vectors remain on their own span

3 [scale=0.7,domain=-2:2] [very thin,color=gray,opacity=0.2] (-2,-2) grid (2,2); [->] (-2,0) – (2,0) node[right] x; [->] (0,-2) – (0,2) node[above] y; [very thick,->,color=red] (0,0)–(1,0) node[below]$\hat{i}$; [very thick,->,color=blue] (0,0)–(-1,0) node[below]α; [decorate,decoration=brace,raise=5pt,blue] (0,0)–(-1,0); at (-0.5,-1) 1; [->,color=red] (0,0)–(0,1) node[left]$\hat{j}$; [color=yellow,opacity=0.2] (0,0) – (1,0) – (1,1) – (0,1);

linearly transformed.

$$\rightarrow \times \begin{bmatrix} 3 & 1 \\ 0 & 2 \end{bmatrix} \rightarrow$$

le=0.5,domain=-2:2] [very thin,color=gray,opacity=0.2] (-4,-4) grid (4,4); [very thin,color=gray,opacity=0.2] (-4,0) – (4,0) ; [very thin,color=gray,opacity=0.2] (-4,2) – (4,2) ; [very thin,color=gray,opacity=0.2] (-4,4) – (4,4) ; [very



Vectors remain on their own span

3 [scale=0.7,domain=-2:2] [very thin,color=gray,opacity=0.2] (-2,-2) grid (2,2); [->] (-2,0) – (2,0) node[right] x; [->] (0,-2) – (0,2) node[above] y;
[very thick,->,color=brown] (0,0)–(1,-1) node[left]γ; [very thick,->,color=purple] (-1,1)–(-2,2) node[left]θ; [very thick,->,color=gray] (1,-1)–(2,-2) node[left]ω;
[very thick,->,color=blue] (0,0)–(-1,1) node[left]β;
[decorate,decoration=brace,raise=5pt,blue] (0,0)–(-1,1); at (-1,0) 1;
[color=yellow,opacity=0.2] (0,0) – (1,0) – (1,1) – (0,1);

linearly transformed.

$$\rightarrow \times \begin{bmatrix} 3 & 1 \\ 0 & 2 \end{bmatrix} \rightarrow$$



le=0.5,domain=-2:2] [very thin,color=gray,opacity=0.2] (-4,-4) grid (4,4);

Eigenvalue & Eigenvector

2 [scale=0.5,domain=-2:2] [very thin,color=gray,opacity=0.2]
(-4,-4) grid (4,4); [very thin,color=gray,opacity=0.2] (-4,0) – (4,0)
; [very thin,color=gray,opacity=0.2] (-4,2) – (4,2) ; [very
thin,color=gray,opacity=0.2] (-4,4) – (4,4) ; [very
thin,color=gray,opacity=0.2] (-4,-2) – (4,-2) ; [very
thin,color=gray,opacity=0.2] (-4,-4) – (4,-4) ;
[very thin,color=gray,opacity=0.2] (-4,-0) – (-2,4) ; [very
thin,color=gray,opacity=0.2] (-4,-4) – (0,4) ; [very
thin,color=gray,opacity=0.2] (-2,-4) – (2,4) ; [very
thin,color=gray,opacity=0.2] (0,-4) – (4,4) ; [very
thin,color=gray,opacity=0.2] (2,-4) – (4,0) ;
[->,opacity=0.2] (-4,0) – (4,0) node[right] x; [->,opacity=0.2]
(-2,-4) – (2,4) node[above] y;
[very thick,->,color=brown] (0,0)–(2,-2) node[left]; [very
thick,->,color=purple] (-2,2)–(-4,4) node[left]; [very
thick,->,color=gray] (2,-2)–(4,-4) node[left];
[very thick,->,color=blue] (0,0)–(-2,2) node[left]β;
[decorate,decoration=brace,mirror,raise=5pt,blue] (0,0)–(-2,2); at



Eigenvalue & Eigenvector

2 [scale=0.5,domain=-2:2] [very thin,color=gray,opacity=0.2] (-4,-4) grid (4,4); [very thin,color=gray,opacity=0.2] (-4,0) – (4,0) ; [very thin,color=gray,opacity=0.2] (-4,2) – (4,2) ; [very thin,color=gray,opacity=0.2] (-4,4) – (4,4) ; [very thin,color=gray,opacity=0.2] (-4,-2) – (4,-2) ; [very thin,color=gray,opacity=0.2] (-4,-4) – (4,-4) ; [very thin,color=gray,opacity=0.2] (-4,-0) – (-2,4) ; [very thin,color=gray,opacity=0.2] (-4,-4) – (0,4) ; [very thin,color=gray,opacity=0.2] (-2,-4) – (2,4) ; [very thin,color=gray,opacity=0.2] (0,-4) – (4,4) ; [very thin,color=gray,opacity=0.2] (2,-4) – (4,0) ; [->,opacity=0.2] (-4,0) – (4,0) node[right] x; [->,opacity=0.2] (-2,-4) – (2,4) node[above] y; [very thick,->,color=brown] (0,0)–(2,-2) node[left]; [very thick,->,color=purple] (-2,2)–(-4,4) node[left]; [very thick,->,color=gray] (2,-2)–(4,-4) node[left]; [very thick,->,color=blue] (0,0)–(-2,2) node[left] $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$;



Eigenvalue & Eigenvector

So maybe you can tell why we can get eigenvalue of matrix from this equation:

$$Ax = \lambda x$$



CONTENT

Vector in Geometry

Determinant in Geometry

Eigenvalue in Geometry

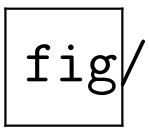
Refference



Refference

2

- Introduction to Linear Algebra(Strang)

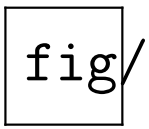




Refference

2

- Introduction to Linear Algebra(Strang)
- Essense of Linear Algebra @3Blue1Brown





Refference

2

- Introduction to Linear Algebra(Strang)
- Essense of Linear Algebra @3Blue1Brown
- Linear algebra and its applications 4th

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Acknowledgements

Thank you for listening!

